
Privacy Analysis of Augmented Reality Glasses

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Weekly Report 11

Michelle Jessica Bösigler
michellej.boesiger@unibas.ch

Department of Mathematics and Computer Science
University of Basel

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Week 11

What has been done (Previous Week)

- Finished and uploaded the code used to generate the images for the network traffic.
- Upload the generated images and analysis generated for the network traffic (for the trial experiments).
- Added documentation for the observations on the images.
- From the meeting, it is noted that the code needs to be changed, so I will need to redo the images / analysis / documentation this week.
- Successfully unlocked the bootloader.
- Received an SD card and was able to successfully root the POCO X5 5G phone. The phone now has the app 'Magisk' which allows the user to manage root permissions.
- Added documentation to the process of rooting the phone.
- Wrote documentation on the methods used to try and bypass certificate pinning for the apk.

What will be done next week (and following weeks as some tasks may extend over multiple weeks)

- This week should first focus on the fact that now that the phone is rooted to continue on:
 - Network traffic: Use mitmproxy to try and collect decrypted network traffic.
 - Bluetooth traffic: Use the root access on the phone to hopefully get potential keys or debug information that could help in decrypting the Bluetooth traffic.
- And when continued, to write documentation on it.
- Update the code, and redo the images and analysis for the Wi-Fi experiments.
- Add documentation for the initial challenges (including twrp and reflashing the phone)

Meeting

Had a meeting with Nima on the 6th, where:

- Discussed about the methods to try and bypass certificate pinning
- Received an SD card to solve for the problem in rooting the phone